# TECHNICAL BULLETIN

Polarix IC-420 High-Ambient Production Performance

| Field | Value | Field | Value |
|---|---|---|---|
| Document ID | TB-002 | Revision | 1 |
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| Document Type | Technical Bulletin | Authority Class | Manufacturer Technical |
| Product Family | Polarix IC Series | Supersedes | None |

## 1. Purpose

Provides additional production-performance guidance for IC-420 under elevated ambient-air and inlet-water temperatures. It does not replace the nominal 240 kg/day product rating.

## 2. Representative Performance

| Condition | Ambient Air | Inlet Water | Expected Production |
|---|---|---|---|
| Standard reference | ~21 C | ~15 C | up to nominal 240 kg/day |
| Warm | 27-31 C | 20-24 C | ~215-230 kg/day |
| High ambient | 32-35 C | 20-25 C | ~195-215 kg/day |
| High ambient + warm water | 32-35 C | 26-30 C | ~180-200 kg/day |
| Severe combined | 36-40 C | 26-30 C | ~160-185 kg/day |

## 3. Interpretation

**Nominal rating is a standardized reference, not guaranteed customer-site output.**

Actual production can vary with ambient temperature, inlet-water temperature, ventilation, maintenance, pressure/flow, and operating conditions.

## 4. Suitability Assessment

• Establish normal and peak daily demand.

• Identify material ambient and inlet-water conditions.

• Use the applicable representative range when elevated conditions are known.

• Compare expected production with demand and required operating margin.

• Evaluate storage strategy separately.

• Do not promise 240 kg/day as guaranteed site output.

## 5. Unknown Conditions

Missing environmental information becomes decision-relevant when customer demand is close enough to IC-420 capability that the recommendation could change. In that case, production suitability remains unconfirmed until the relevant conditions are established.

## 6. Relationship to PT-001

PT-001 remains authoritative for the nominal 240 kg/day rating. TB-002 supplies contextual performance evidence. Different values are not a conflict when they describe different operating conditions.

## 7. Related Documentation

• PT-001 - Polarix IC Series Product Catalog

• PT-002 - Polarix IC Series Installation Guide

- CP-001 - AquaPure Equipment Compatibility Matrix
- CP-003 - Polarix Ice System Component Compatibility Guide

*Controlled technical knowledge. Apply only to the documented product population, relationship, and conditions.*